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Subject: Application for the M2 Internship – Innovative Electrochemical Devices Based on Liquid-Liquid Microinterfaces for Detecting Water Pollutants — NIMBE/LEDNA, CEA Paris-Saclay

Dear Ms. Cannizzo and Ms. Peulon-Pagé,

I am a final-year Industrial Chemical Engineering student at INSAT (Université de Carthage, Tunisia), specializing in Process and Environment, and I am writing to apply for the M2 internship on electrochemical devices based on liquid-liquid microinterfaces for detecting water pollutants at NIMBE/LEDNA. I am available from 1 February 2027, for 6 months (3 to 9 months possible).

My coursework has given me a solid grounding in electrochemical kinetics and electrochemical measurements, which I put into practice in an academic project on the electrochemical treatment of an industrial effluent by electroflocculation: I recorded current-potential curves and pulse polarography on aluminum electrodes, calculated the faradaic yield with Faraday's law, and measured the removal of Cd2+ and Zn2+ (85% and 57% in 20 min). During my internship at ETAP, I also performed physico-chemical and ionic analyses of water samples by ion chromatography and ICP-AES. This combination of simple electrochemical methods and water analysis is directly analogous to the measurement and detection-condition work your internship describes.

Your laboratory's approach, enhancing the selectivity of liquid-liquid microinterface sensors with cage molecules and nano-objects, is exactly the kind of applied electrochemistry I want to build my final-year project around, with a clear purpose: monitoring pollutants in different types of water. Cage molecules, nano-object characterization and your electrochemical control software are new to me specifically, but my internship at SEGULA Technologies, where I built and recalibrated a Li-ion cell model in MATLAB and reduced its RMSE from 202.13 mV to 40.5 mV, shows that I learn new methods quickly and work independently.

I am drawn to this internship because it sits precisely at the intersection of electrochemistry and water quality, the two threads of my training.

I would welcome the opportunity to discuss my application further and am available for an interview at your convenience. Thank you for your consideration.

Sincerely,
Nermine Moumen